10.8 Smoothing

The axis-aligned motions of a grid planner and the randomized motions of sampling planners may lead to jerky motion of a robot. One approach to dealing with this issue is to let the planner handle the work of searching globally for a solution, then post-process the resulting motion to make it smoother.

There are many ways to do this; two possibilities are outlined below.

法一：非线性优化

Nonlinear Optimization While gradient-based nonlinear optimization may fail to find a solution if initialized with a random initial trajectory, it can make an effective post-processing step, since the plan initializes the optimization with a "reasonable" solution. The initial motion must be converted to a parametrized representation of the controls, and the cost $J(u(t), q(t), T)$ can be expressed as a function of $u(t)$ or $q(t)$. For example, the cost function

代价函数

$$J = \frac{1}{2}\int_0^T \dot{u}^{\mathrm{T}}(t)\dot{u}(t)dt$$

$\dot{u}$ 控制量变化率

力矩高频更可能坏掉

penalizes rapidly changing controls. This has an analogy in human motor control, where the smoothness of human arm motions has been attributed to minimization of the rate of change of torques at the joints [188].

法二：细分与重连

法① 随机选点重连

Subdivide and Reconnect A local planner can be used to attempt a connection between two distant points on a path. If this new connection is collision-free, it replaces the original path segment. Since the local planner is designed to produce short, smooth, paths, the new path is likely to be shorter and smoother than the original. This test-and-replace procedure can be applied iteratively to randomly chosen points on the path. Another possibility is to use a recursive procedure that subdivides the path first into two pieces and attempts to replace each piece with a shorter path; then, if either portion cannot be replaced by a shorter path, it subdivides again; and so on.

RRT 路径平滑算法

Shortcut

(与RRT*思想相同)

树状结构中重新连

法② 分治思想、(递归)

先看这边能否连

不能再二分……递归下去

整条路找到后，一次性处理

10.9 Summary

- A fairly general statement of the motion planning problem is as follows. Given an initial state $x(0) = x_{\text{start}}$ and a desired final state x_{goal}, find a time T and a set of controls $u : [0, T] \to \mathcal{U}$ such that the motion satisfies $x(T) \in \mathcal{X}_{\text{goal}}$ and $q(x(t)) \in \mathcal{C}_{\text{free}}$ for all $t \in [0, T]$.
- Motion planning problems can be classified in the following categories: path planning versus motion planning; fully actuated versus constrained